

When Are Two Conformers the Same?

Symmetry, Chirality, and Conformer De-duplication

Notes for ChemDM

A conceptual reference for the conformer-identity question that drives the clustering in `src/chemdm/Cluster.py`. It captures *why* two distinct-looking 3D structures can be the same chemical conformer, why plain atom-relabelling cannot detect this, and the exact rule — gate a reflection on molecular achirality — that resolves it. The running example is nitrosocyclopentane, O=NC1CCCC1, whose pipeline output first raised the question.

1 The question

Sample and relax the conformers of O=NC1CCCC1. Three survive RMSD clustering, two of them at *exactly* equal energy:

conformer	energy (kJ/mol)	
0	-58265.3736	} degenerate
1	-58265.3735	
2	-58263.1315	distinct pucker

Conformers 0 and 1 are *not* superimposable by any rotation — their heavy-atom RMSD after optimal proper alignment is 0.53 Å. Yet they have identical energy. Are they two conformers or one? The answer requires being precise about what “the same conformer” means.

2 A conformer is an equivalence class

The potential energy surface (PES) $E : \mathbb{R}^{3N} \rightarrow \mathbb{R}$ assigns an energy to every labelled geometry $X = (x_1, \dots, x_N)$, $x_i \in \mathbb{R}^3$. A *conformer* is a local minimum of E — but not a single point. Physics is blind to:

- **Rigid motion.** Translating or rotating the whole molecule changes no energy. So X and $RX + t$ are the same, for $R \in \text{SO}(3)$, $t \in \mathbb{R}^3$.
- **Relabelling identical atoms.** If a permutation π of the atoms maps the molecular graph onto itself (a *graph automorphism*) — swapping two chemically equivalent hydrogens, say — then X and πX describe the same substance.

So a conformer is really an *equivalence class* of geometries under the group generated by rigid motions and the molecule’s symmetry operations. The whole question “are 0 and 1 the same?” is the question “does the group contain an operation taking one to the other?”. Everything below is about which operations belong in that group.

3 Proper vs. improper: the handedness invariant

The isometries of $\mathbb{R}^3$ form the group $O(3)$, which splits into two pieces by the determinant of the linear part:

$$\det = +1 : \text{ proper (rotations, } SO(3)), \quad \det = -1 : \text{ improper (reflections, inversions, } S_n).$$

A proper operation preserves *handedness*; an improper one reverses it. Handedness (chirality) of a rigid point cloud is a genuine invariant: no rotation, and no relabelling, can turn a left hand into a right hand.

This is the crux. A graph automorphism π is a *permutation* — it reorders labels, it does not act on space. Composing π with a proper rotation is still an orientation-preserving operation overall. Therefore:

Relabelling plus rotation can never map a structure onto its mirror image.

If conformers 0 and 1 are mirror images, no amount of clever atom-matching will superimpose them. To detect their identity you must *explicitly* include a reflection — and whether you are *allowed* to is exactly the chirality question.

4 The symmetry-corrected RMSD

Fix two heavy-atom coordinate sets $A, B \in \mathbb{R}^{H \times 3}$ (hydrogens excluded; they follow their heavy neighbours). Let $\mathcal{P}$ be the heavy-atom graph automorphisms of the molecule, computed once from the 2D graph. Define

$$\text{RMSD}^{\text{prop}}(A, B) = \min_{\pi \in \mathcal{P}} \min_{R \in SO(3), t} \left[\frac{1}{H} \sum_i \|Ra_i + t - b_{\pi(i)}\|^2 \right]^{1/2}.$$

The inner minimisation over (R, t) is solved in closed form by the *Kabsch* algorithm, using the *proper* variant (a determinant-sign correction forces $\det R = +1$, forbidding the reflected fit). This is `kabsch_align_numpy` in `src/chemdm/geometry.py`, and `_symmetry_corrected_heavy_rmsd` in `Cluster.py` is exactly the min over $\mathcal{P}$ above.

Using the proper Kabsch is a deliberate choice: it *protects chirality*. Two genuine enantiomers of a chiral molecule must never be merged, and $\text{RMSD}^{\text{prop}}$ keeps them apart because it cannot cross the mirror. The cost is that it *also* keeps mirror-image conformers of an *achiral* molecule apart — which is the over-splitting we are trying to fix.

5 Chirality decides whether reflection is a symmetry operation

Here is the clean statement that resolves everything.

Reflection belongs to a molecule’s symmetry group **if and only if** the molecule is achiral.

A molecule is *achiral* exactly when it possesses an improper symmetry element — a mirror plane σ , an inversion centre i , or an improper-rotation axis S_n — i.e. some improper isometry (possibly composed with a relabelling) maps the molecule onto a superimposable copy of itself. That is what “superimposable on its mirror image” means.

- **Achiral molecule.** An improper symmetry operation *exists*. Two geometries related by it are related by a genuine symmetry of the molecule, hence they are *the same conformer*. Reflection is legitimate; include it.
- **Chiral molecule.** No improper symmetry operation exists. Reflection turns the molecule into its *enantiomer* — a different substance. Mirror-image conformers belong to two different molecules and must be kept distinct. Reflection is illegitimate; forbid it.

So the achiral/chiral test is not a heuristic knob. It is the precise condition under which the reflection is a member of the equivalence group. Detect it once from the 2D graph (no stereocentres and no stereo double bonds $\Rightarrow$ achiral) and gate the reflected fit on it.

6 Worked example: the nitrosocyclopentane degeneracy

The nitroso group $\text{O}=\text{N}-\text{C}$ is locally planar. The two degenerate conformers are the cyclopentane ring puckering “up” on one side of that plane and “down” on the other, versus the reverse — mirror images across the $\text{O}-\text{N}-\text{C}$ plane. The molecule is achiral (no stereocentres), so this mirror, composed with the ring-reversal relabelling $\text{C}_2 \leftrightarrow \text{C}_5$, $\text{C}_3 \leftrightarrow \text{C}_4$, is a real σ symmetry element.

Numerically, on the two relaxed conformers ($H = 6$ heavy atoms, only two heavy automorphisms: identity and ring-reversal):

operation applied to align conf. 0 onto conf. 1	heavy RMSD (Å)
proper rotation + identity relabel	0.526
proper rotation + ring-reversal relabel	0.939
reflection + identity relabel	1.037
reflection + ring-reversal relabel	0.000

Neither half alone works: relabelling cannot flip handedness (rows 1–2), and a bare reflection sends the ring atoms across the plane onto the *wrong* labels (row 3). Only the *composite* improper operation — reflection *and* the ring relabelling, i.e. the molecule’s σ element — superimposes them exactly (row 4). That composite is a symmetry operation of this achiral molecule, which is precisely why the two are the same conformer.

7 Are they chemically identical? Yes.

For an isolated molecule in an achiral environment, conformers 0 and 1 are *the same conformer*, counted once. Being related by a molecular symmetry operation, they are indistinguishable by every achiral observable — energy (exactly equal, by symmetry), dipole magnitude, moments of inertia, IR/Raman, NMR in an achiral solvent, and all thermodynamic contributions — and they interconvert by a low-barrier ring flip.

The textbook analogue is **gauche-butane**. The two gauche forms g^+ ($\phi = +60^\circ$) and g^- ($\phi = -60^\circ$) are mirror images. Butane is universally reported as having *two* conformers — anti and gauche — not three: the gauche conformer simply carries a multiplicity of 2. Nitrosocyclopentane is the same story: conformers 0 and 1 are the g^+/g^- of the ring pucker (one conformer, degeneracy 2), and conformer 2 is the genuinely distinct pucker. Our pipeline already reports exactly two conformers for butane itself, which is the sanity check.

The degeneracy is not lost. Merging does not discard information: the mirror pair contributes a *symmetry factor* of 2 to the statistical weight (the conformer’s degeneracy / symmetry number), not a second entry in the list. If Boltzmann populations are needed, carry the multiplicity; the two mirror wells are one state with weight 2.

The one exception. In a *chiral* environment — a chiral solvent or catalyst, enantioselective binding, or circular-dichroism spectroscopy — the two conformational enantiomers respond oppositely and are distinguishable. If the ensemble feeds such a calculation, keep both. For every achiral-observable purpose (energies, populations, ordinary spectra, matching to CREST references) they are redundant and one should be removed.

8 The de-duplication rule, precisely

1. Compute the heavy-atom graph automorphisms $\mathcal{P}$ once from the molecule (identity included).
2. Determine achirality once: no assigned or potential stereocentres and no stereo double bonds.
3. Two relaxed conformers A, B are the same iff, after energy pre-filtering,

$$\min \left(\text{RMSD}^{\text{prop}}(A, B), \underbrace{[\text{achiral}] \cdot \text{RMSD}^{\text{refl}}(A, B)}_{\text{only if achiral}} \right) \leq \tau,$$

where $\text{RMSD}^{\text{refl}}$ is the same minimisation but allowing the reflected Kabsch fit, and $[\text{achiral}]$ is 1 for an achiral molecule and $+\infty$ (disabled) for a chiral one.

For a chiral molecule the second term is switched off and the behaviour is identical to today’s proper-only clustering: genuine enantiomers are never merged. For an achiral molecule the reflected fit is enabled, and it can only ever collapse pairs that a real improper symmetry operation of the molecule connects — exactly the conformational enantiomers, and nothing else. (In the data above, the distinct pucker, conformer 2, stays 0.33 Å from the mirror pair even *with* reflection: it is not over-merged.)

One subtlety worth a footnote. “Achiral molecule” refers to the *configuration* (the 2D graph): no fixed stereocentres. An individual puckered *conformation* may still be geometrically chiral (conf. 0 has C_1 point-group symmetry). That is fine and expected — it is the whole reason the two conformers look different. The molecule being configurationally achiral is what guarantees the two chiral conformations are mirror-related and interconvert, hence the same species.